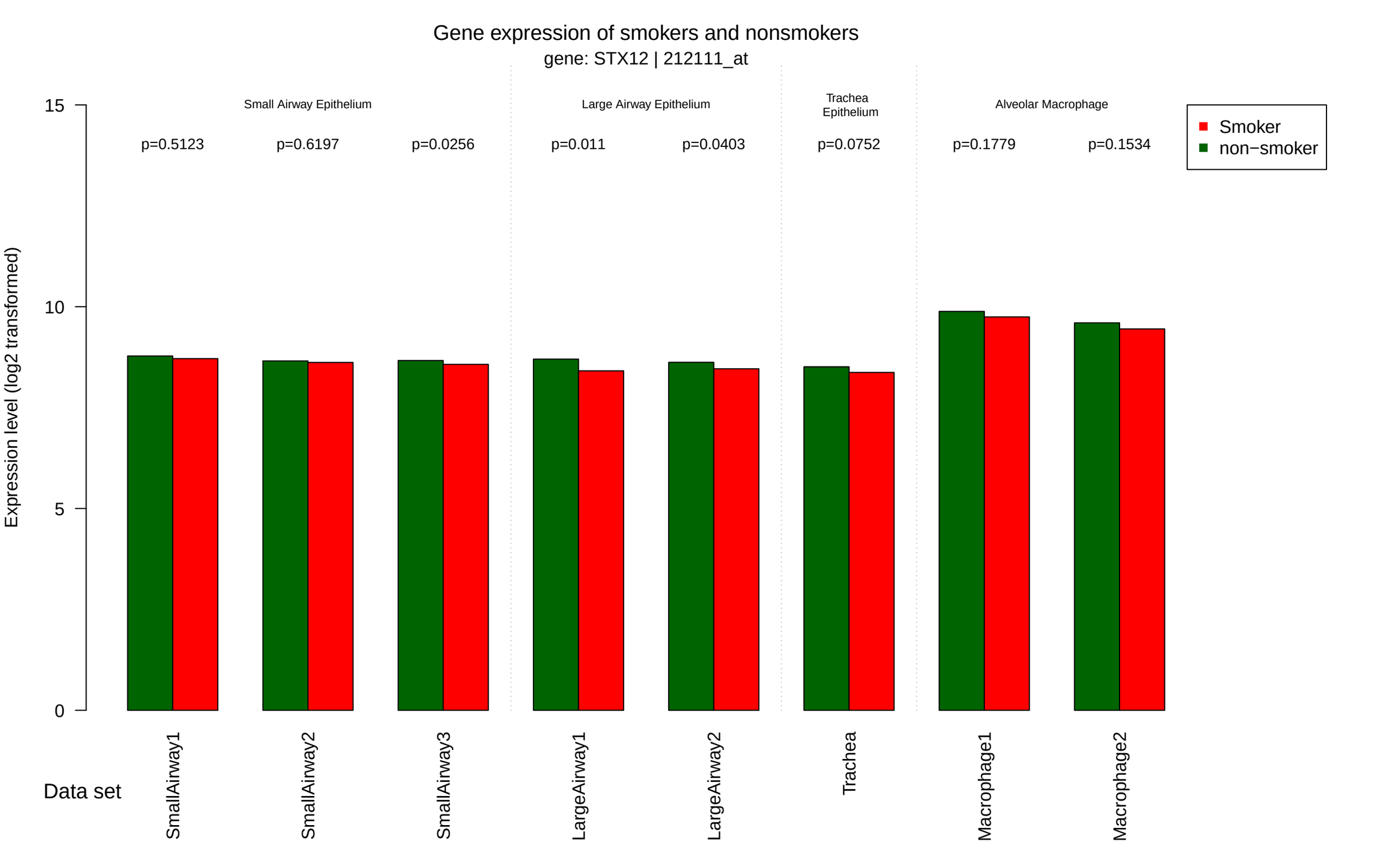
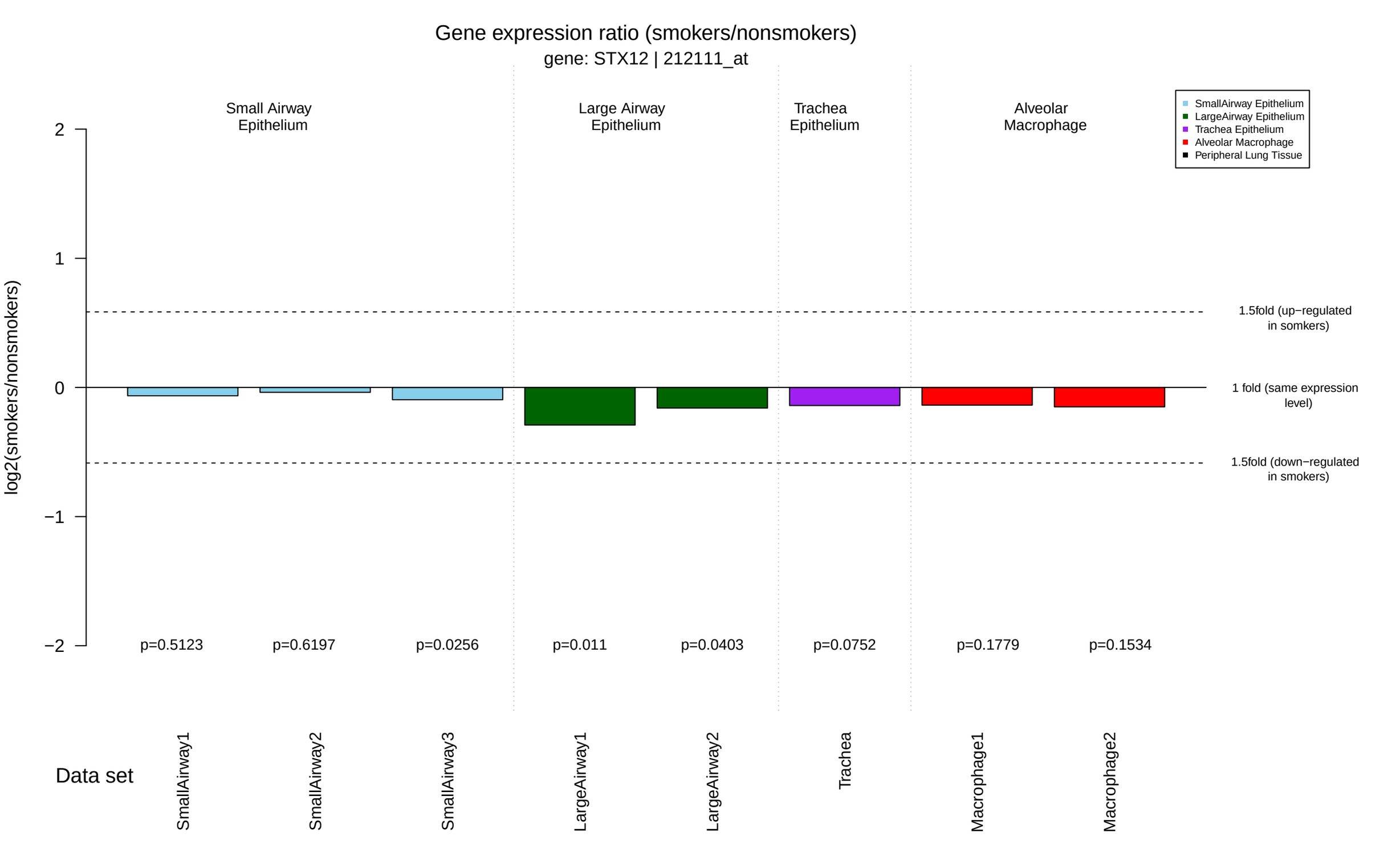
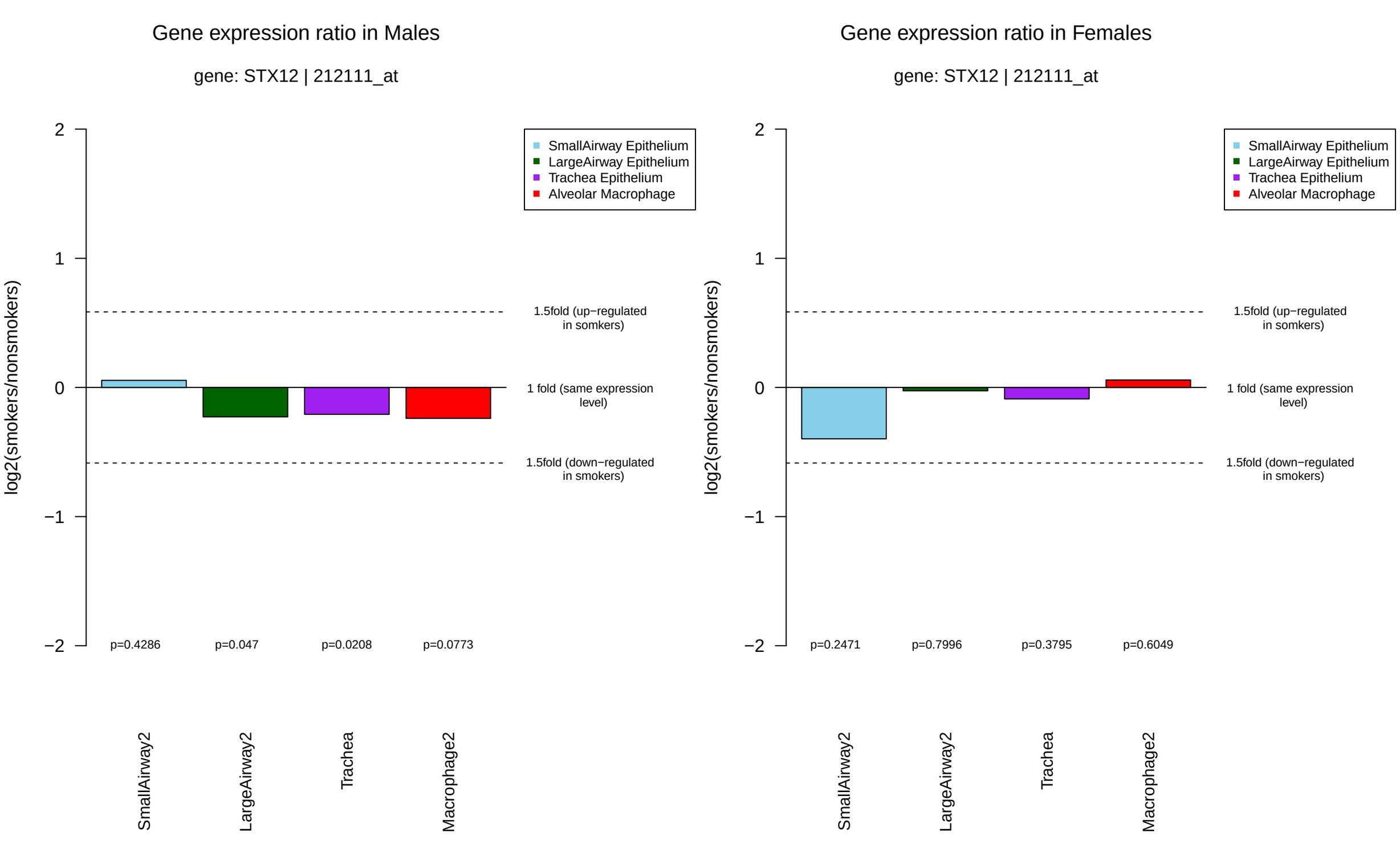
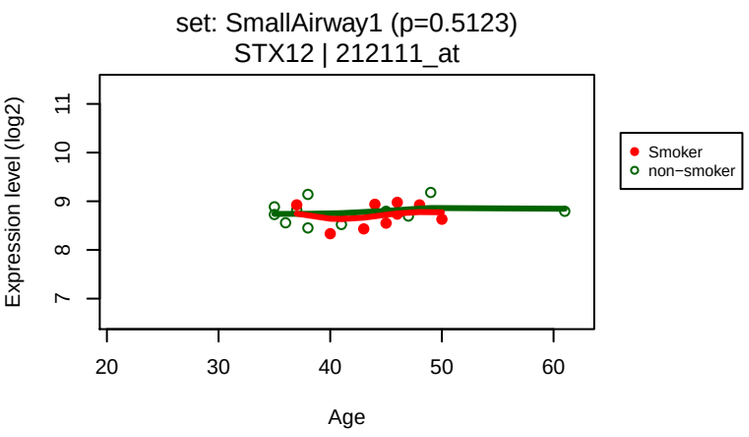
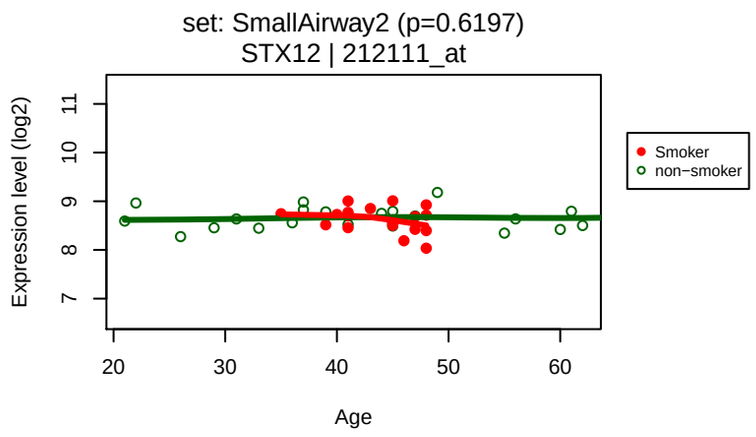
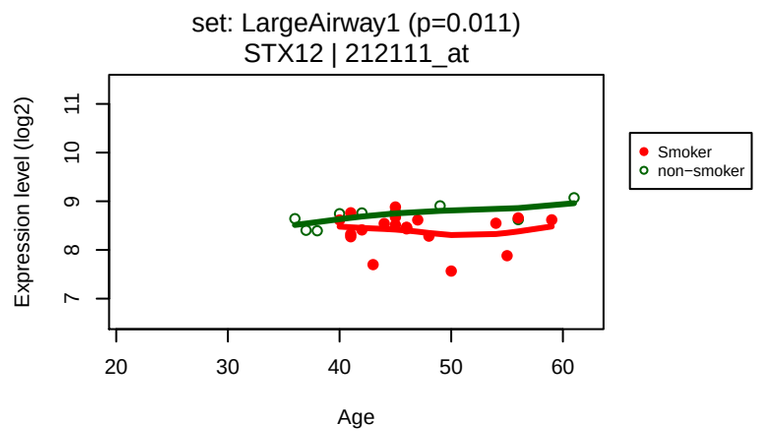

Gene expression of smokers and nonsmokers
gene: STX12 | 212111_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5123
p=0.6197
p=0.0256
p=0.011
p=0.0403
p=0.0752
p=0.1779
p=0.1534
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

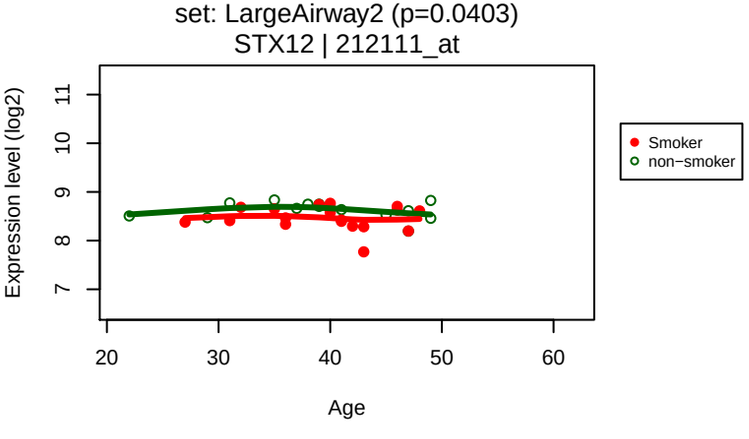

Gene expression ratio (smokers/nonsmokers)
gene: STX12 | 212111_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5123
p=0.6197
p=0.0256
p=0.011
p=0.0403
p=0.0752
p=0.1779
p=0.1534
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

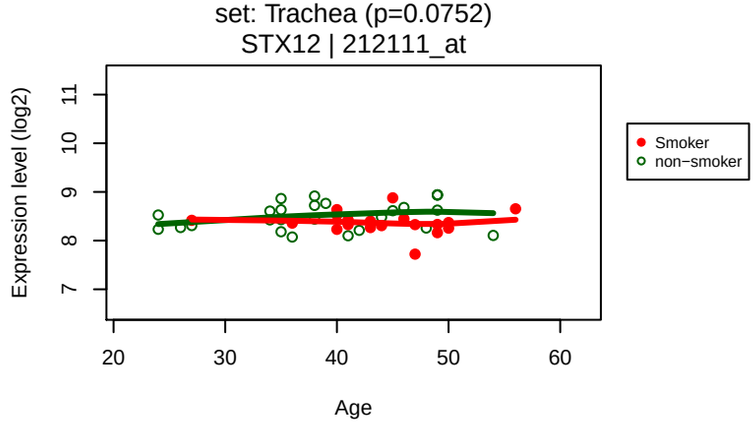

Gene expression ratio in Males
gene: STX12 | 212111_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4286
p=0.047
p=0.0208
p=0.0773
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STX12 | 212111_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2471
p=0.7996
p=0.3795
p=0.6049
SmallAirway2
LargeAirway2
Trachea
Macrophage2

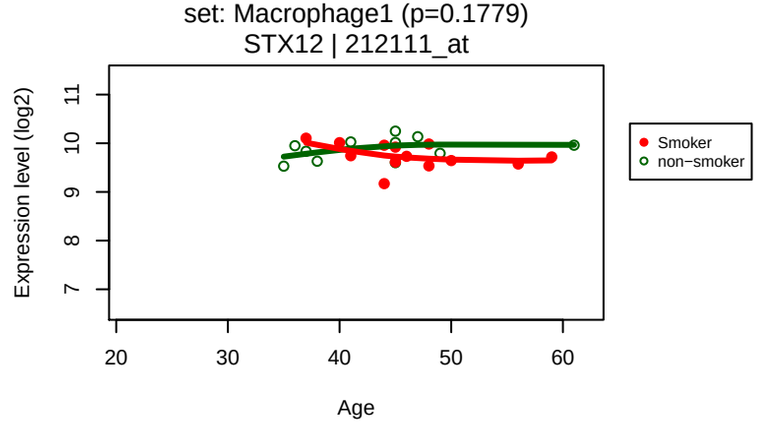

set: SmallAirway1 (p=0.5123)
STX12 | 212111_at
Expression level (log2)
Age
Smoker
non−smoker

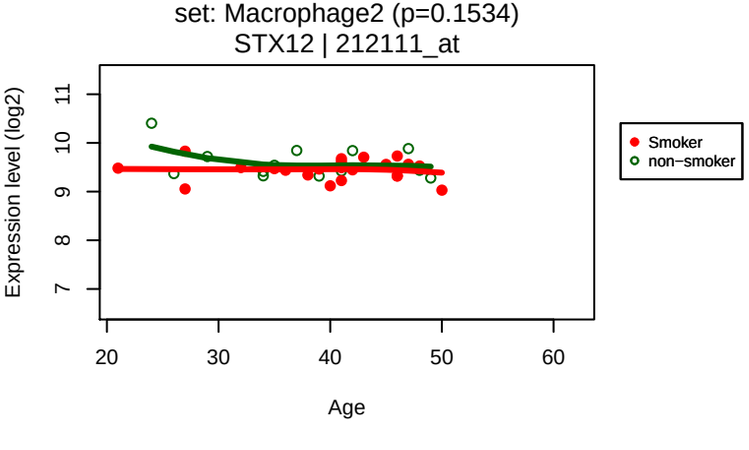

set: SmallAirway2 (p=0.6197)
STX12 | 212111_at
Expression level (log2)
Age
Smoker
non−smoker

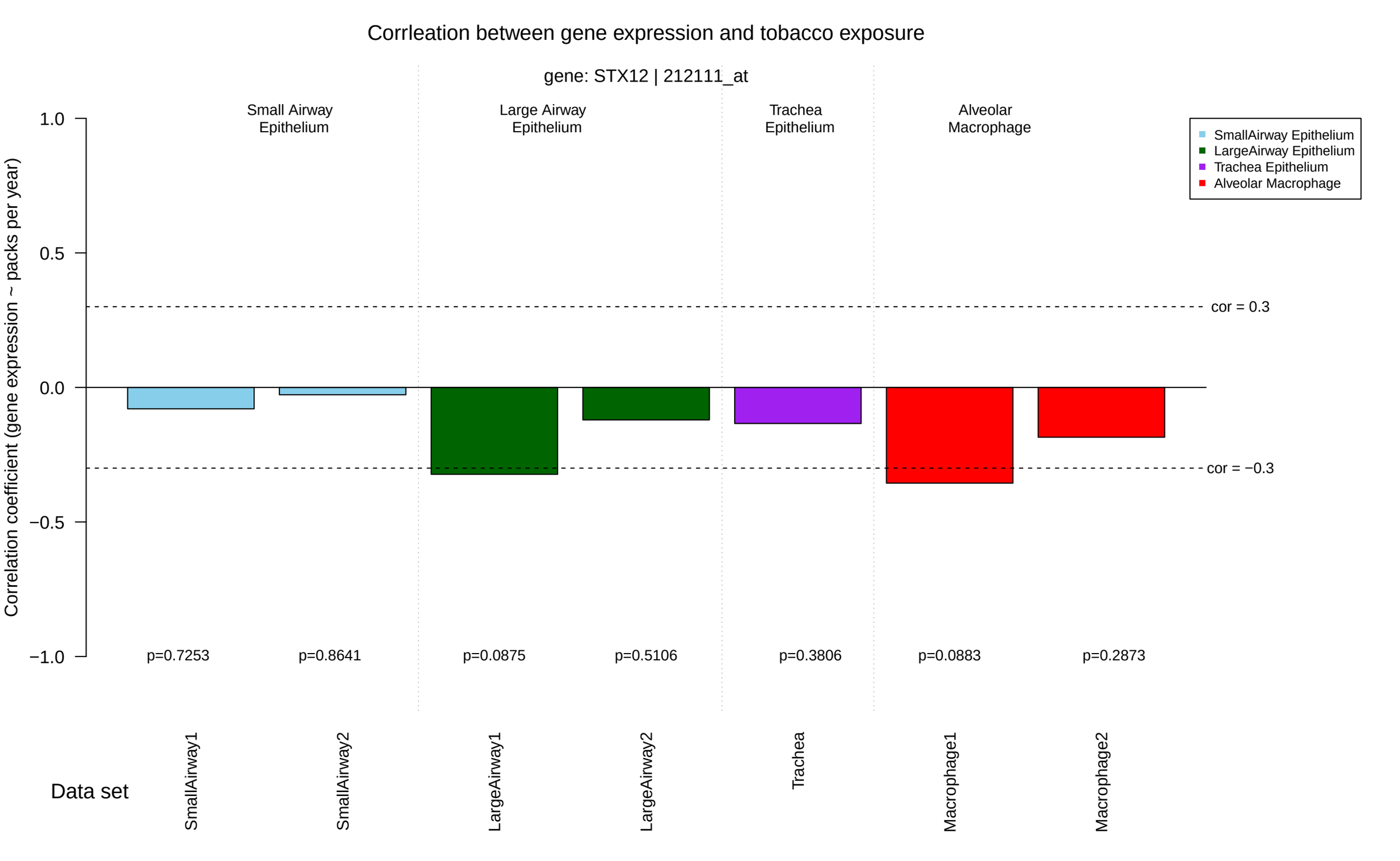

set: LargeAirway1 (p=0.011)
STX12 | 212111_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0403)
STX12 | 212111_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0752)
STX12 | 212111_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1779)
STX12 | 212111_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1534)
STX12 | 212111_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STX12 | 212111_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7253
p=0.8641
p=0.0875
p=0.5106
p=0.3806
p=0.0883
p=0.2873
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2